

Entity Relationship Diagrams

Exercise 2

1. Explain the distinctions among the terms primary key, candidate key, and super key.
2. Construct an E-R diagram for a car insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents.
3. Construct an ER diagram for the hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted.
4. A university registrar's office maintains data about the following entities: (a) courses, including number, title, credits, syllabus, and prerequisites;(b) course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom;(c) students, including student-id, name, and program; and(d)instructors, including identification number, name, department, and title. Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modelled. Construct an E-R diagram for the registrar's office. Document all assumptions that you make about the mapping constraints.